

Assignment 4 Insights Report

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Executive Summary

This report presents an analysis of the LAPD Crime Dataset to identify factors influencing whether a criminal case gets solved and to build predictive models for case resolution. The dataset is significantly imbalanced — only 20.9% of cases are solved — which was addressed through class weighting in both models.

Key findings show that weapon usage is the strongest predictor of case solvability: crimes involving bladed weapons or strong-arm force are resolved at higher rates than those involving firearms. Victim demographics play a secondary role, with female victims having a modestly higher solve rate than male victims. Geographically, solve rates across Los Angeles neighborhoods are consistently low, ranging between 20–30% across most areas.

Two classification models were trained and evaluated: Logistic Regression and Random Forest. Both achieved similar overall accuracy. The Logistic Regression model showed stronger recall for solved cases, making it the preferred model when the goal is correctly identifying solvable cases. The Random Forest model performed better at identifying unsolved cases but struggled with the minority class.

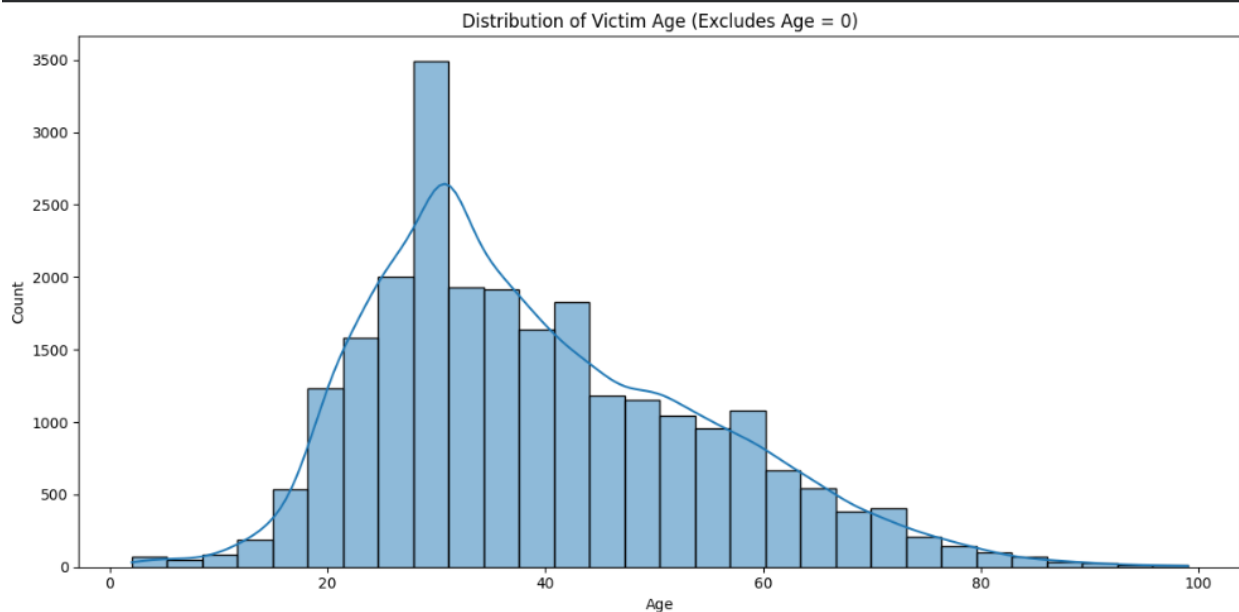
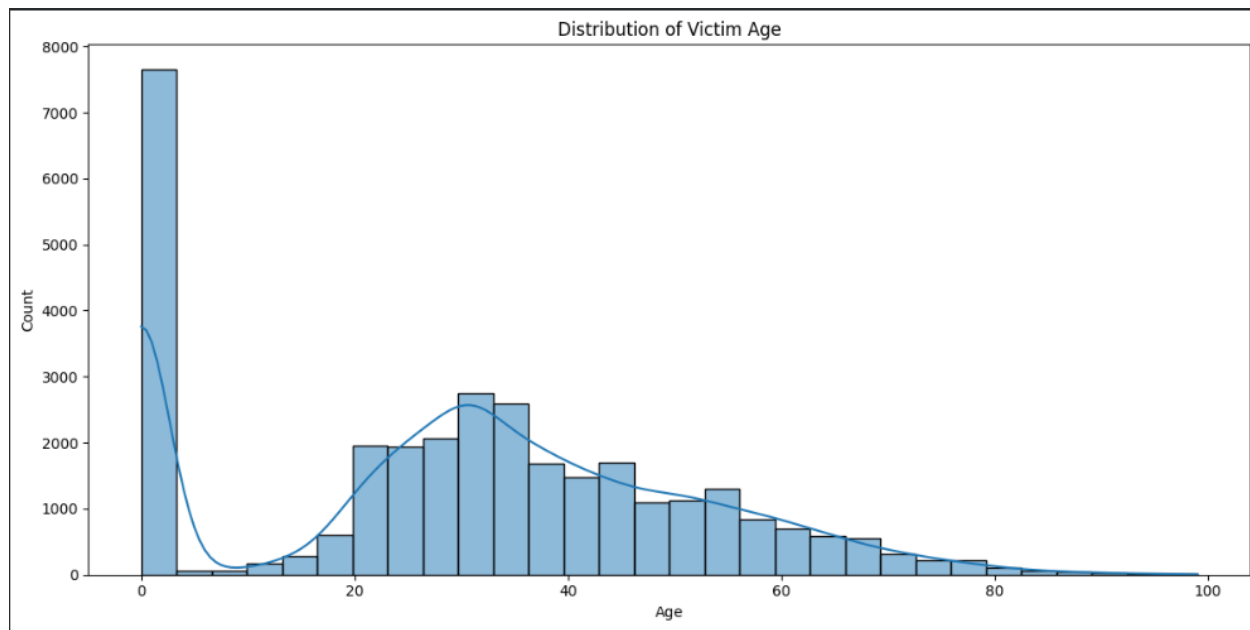
Exploratory Analysis Overview

Data Cleaning

The dataset was cleaned through the following steps:

- Date columns (date_reported, date_occurred) were moved to datetime format
- Inconsistent gender values were normalized: M → Male, F → Female, X/H → Other
- Four columns with excessive missing values and no analytical value were dropped: crime_code_2, crime_code_3, crime_code_4, and cross_street
- Missing victim_age values were imputed using the median (as an integer). Age = 0 represents crimes where a victim's age was not recorded — primarily vehicle and bike theft — and is preserved as a distinct group in the age_group feature
- Missing victim_gender and victim_ethnicity were filled with 'Unknown'; missing weapon_code was filled with 0

The victim age distribution was visualized both with and without zero-age records. Excluding them reveals a right-skewed distribution centered around 30 years old.

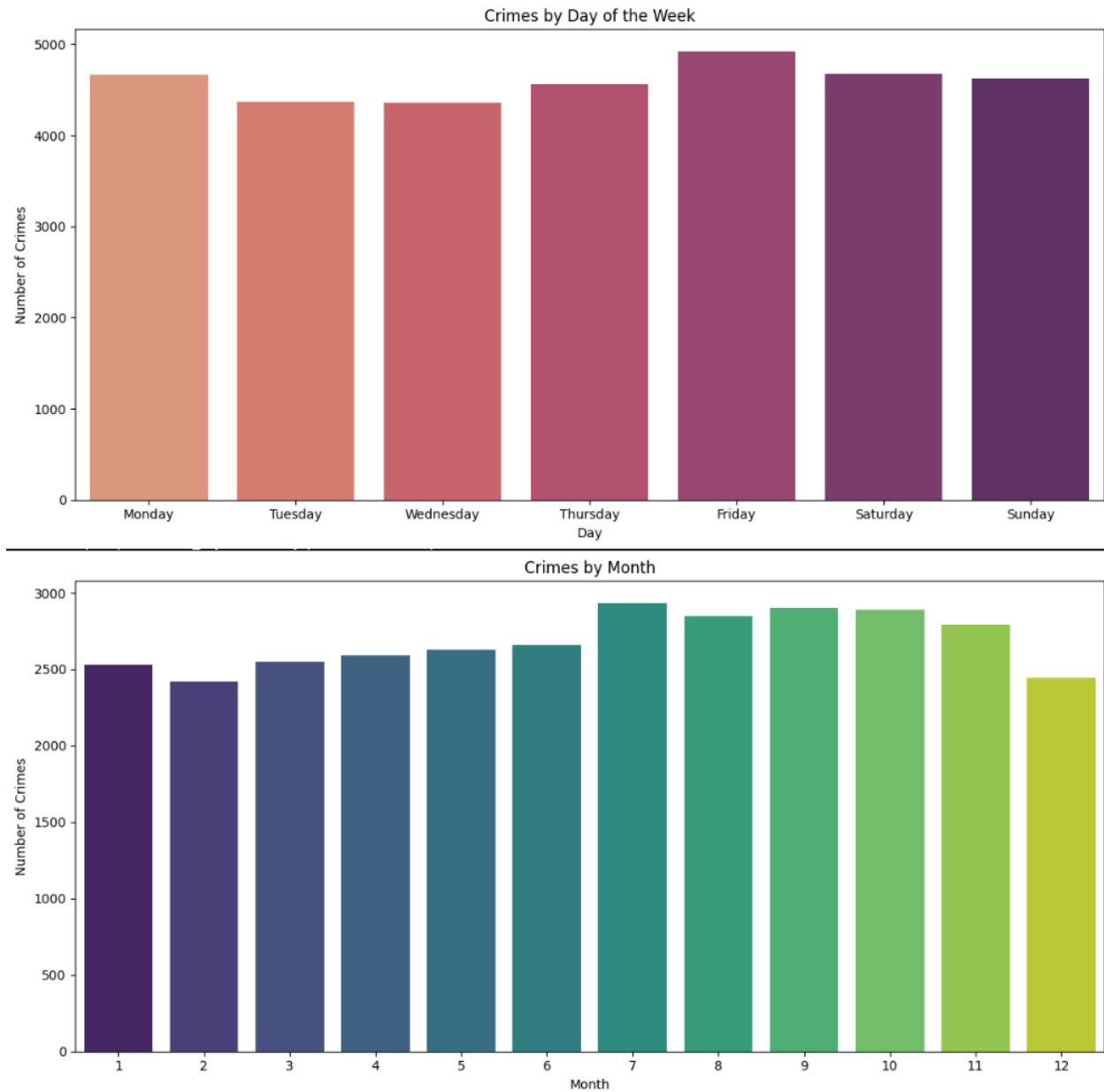


Class Imbalance

The target variable (`case_solved`) is highly imbalanced: 160,650 unsolved cases (79.1%) versus 42,439 solved cases (20.9%). This imbalance was explicitly addressed by using `class_weight='balanced'` in both models, which adjusts the loss function to penalize misclassification of the minority (solved) class more heavily.

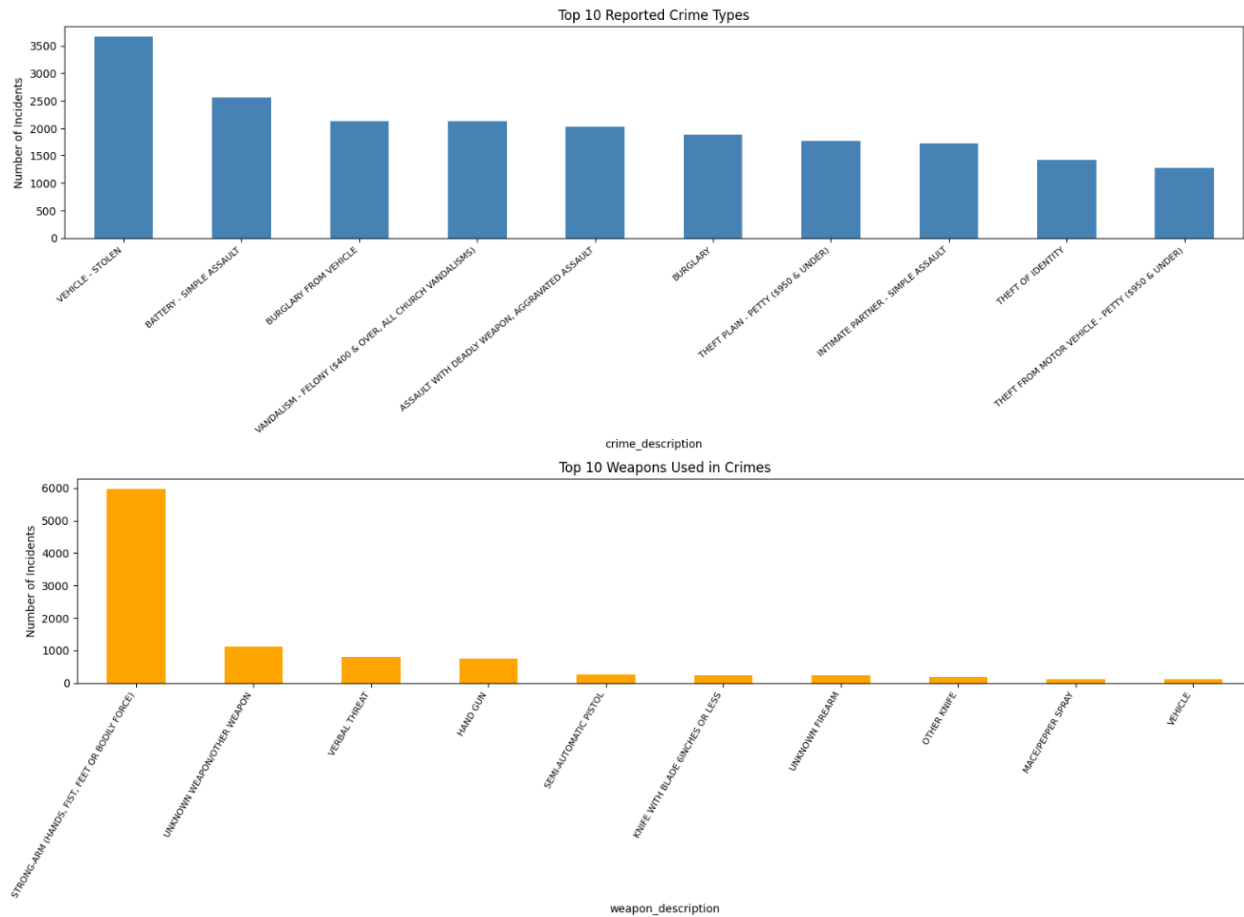
Temporal Patterns

Crimes are fairly evenly distributed across months with a small increase in summer and fall months. By day of the week, Friday shows a slight peak in incident frequency, though no single day stands out dramatically. The hour feature captures intra-day patterns that are more informative for the models.



Crime and Weapon Distribution

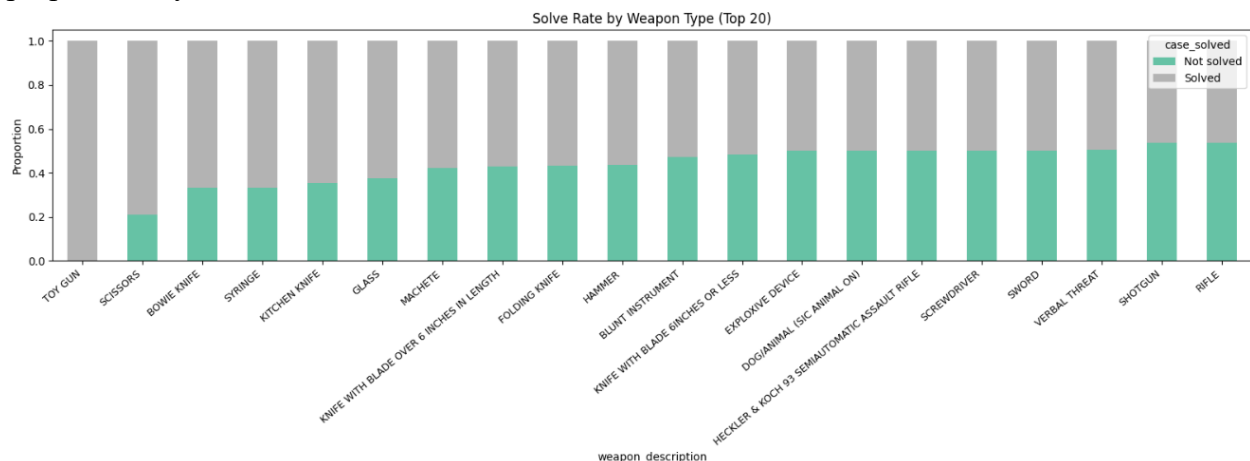
Vehicle theft or Grand Theft Auto is the most reported crime, followed by Battery, Simple Assault and Vandalism. For weapons, physical force (strong-arm) leads by a large margin over all other weapon types, followed by unknown/other and verbal threats. This indicates that the majority of crimes in the dataset do not involve firearms or bladed weapons.



Influential Factors on Case Solvability

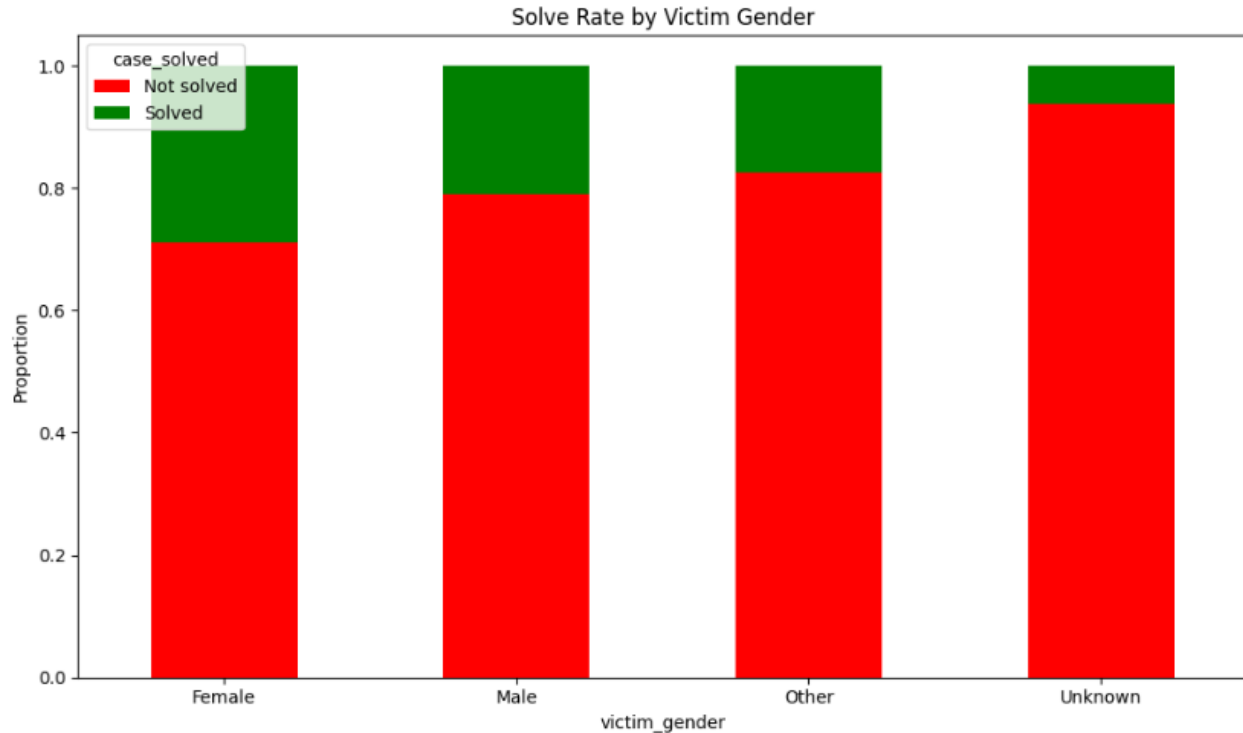
Weapon Usage

Weapon usage is the most influential factor on case solvability. Crimes involving bladed weapons show the highest solve rates, while firearms result in a more even split between solved and unsolved. Crimes with no recorded weapon — the majority — have the lowest solve rates proportionally.



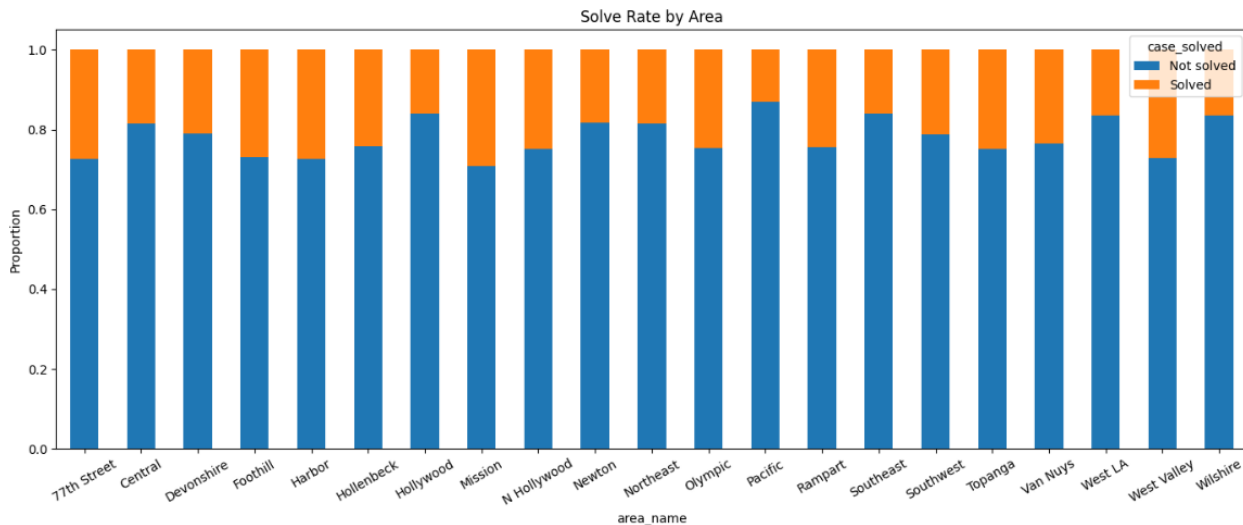
Victim Gender

Female victims have a modestly higher solve rate than male victims. Cases where gender is unknown are almost never solved, likely because these records correspond to property crimes with no direct victim like a vehicle theft.



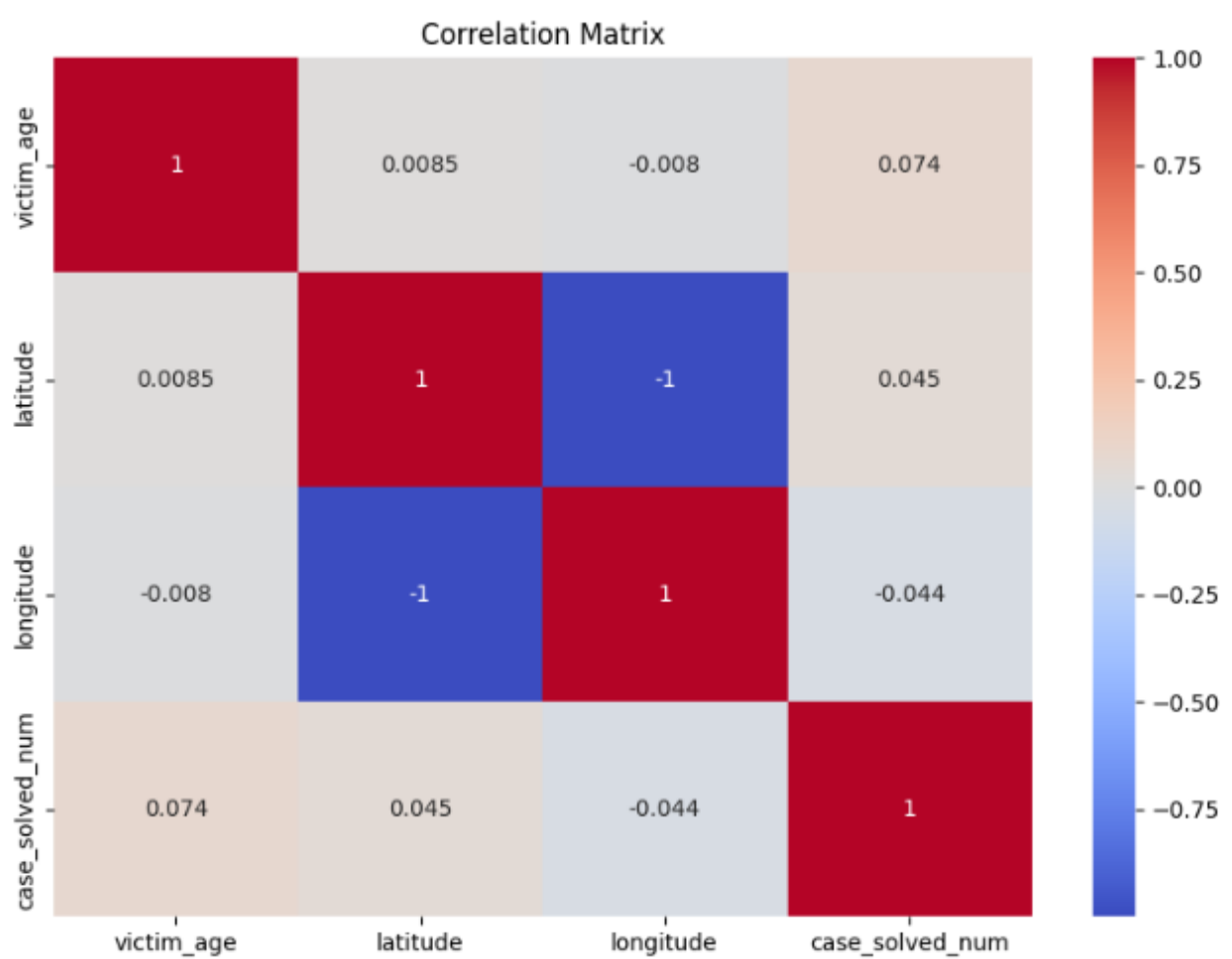
Geographic Distribution

Solve rates across Los Angeles neighborhoods are uniformly low, with most areas resolving between 20% and 30% of reported incidents. Mission, North Hollywood, and West Valley show slightly higher rates, but no area demonstrates a dramatically different pattern. This suggests that geographic location alone is a weak predictor of solvability.



Correlation Analysis

The correlation heatmap reveals that no single numeric feature strongly correlates with case solvability on its own. Weapon usage shows the most signal among the numeric variables. This confirms that categorical features (crime type, area, gender) and engineered features (age group, crime category, hour) carry most of the predictive value in this dataset.



Model Building and Feature Engineering

Feature Engineering

The following features were engineered or derived to improve model performance:

- hour: hour of day extracted from date_occurred — captures intra-day crime patterns
- is_weekend: binary flag for Saturday/Sunday
- weapon_used: binary flag — 1 if any weapon was recorded, 0 otherwise
- age_group: categorical bins (Non-Person, Young, Adult, Senior, Elderly) — separates property crimes where age = 0 from actual victim age groups
- crime_category: non-overlapping classification of crime codes into Violent, Property, Public Order, or Other — based on the actual crime codes present in the lookup file

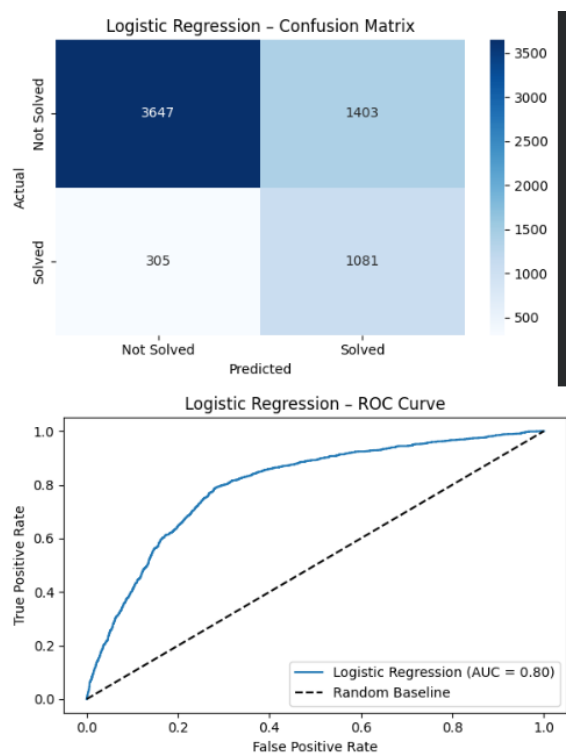
Numeric features (hour, month, latitude, longitude, is_weekend, weapon_used) were standardized using StandardScaler. Categorical features were encoded using OneHotEncoder with unknown category handling. Binary features (is_weekend, weapon_used) were kept as numeric to avoid unnecessary expansion of the feature matrix.

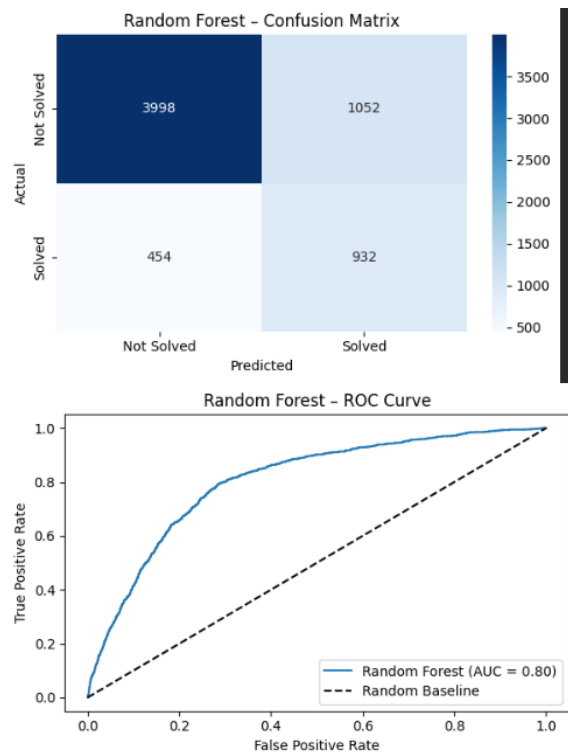
The preprocessor was defined separately but fitted exclusively inside the sklearn Pipeline on training data only, preventing data leakage into the test set.

Model Results

Both models were evaluated on a stratified 80/20 train/test split (random_state=42). Update the table below with your model output values after running the notebook:

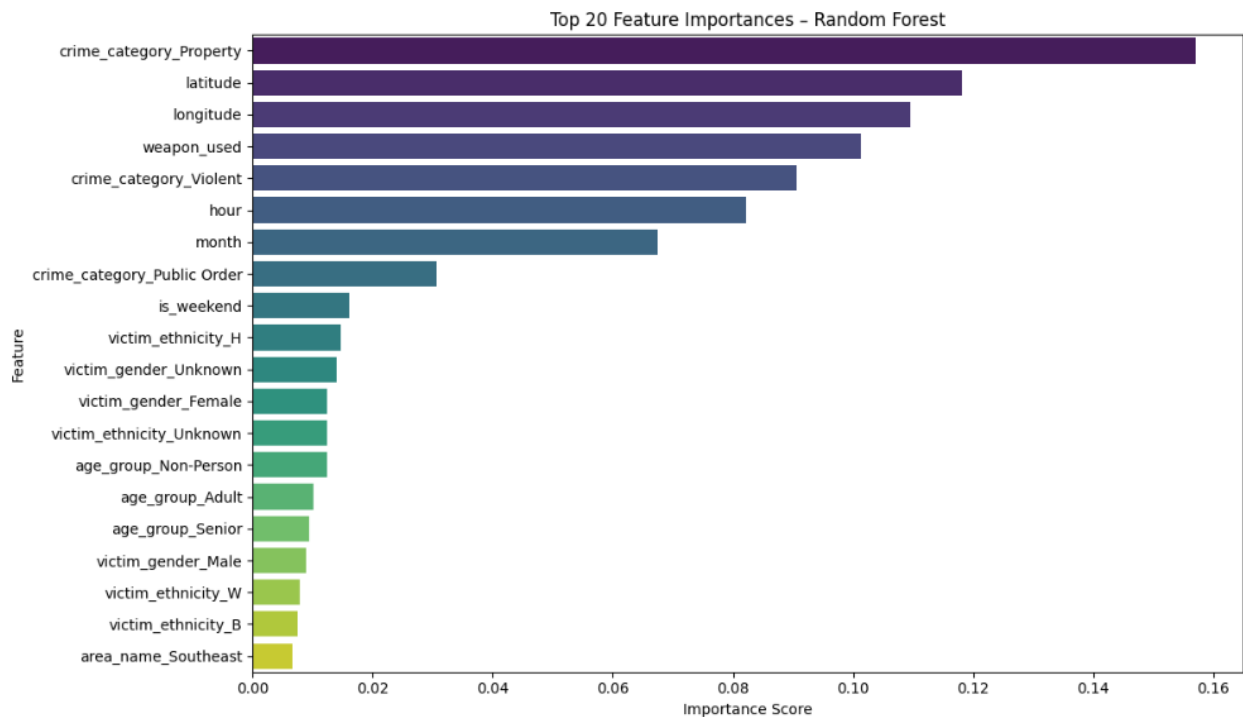
Metric	Logistic Regression	Random Forest
Accuracy	0.73	0.77
Precision	0.44	0.47
Recall	0.78	0.67
F1-Score	0.56	0.55
AUC-ROC	0.80	0.80





Feature Importance

The Random Forest model provides feature importance scores that identify which variables most influenced predictions. The top features are expected to include `weapon_used`, `hour`, geographic coordinates (latitude/longitude), and `crime_category`.



Feature importance confirms that weapon usage and time-of-occurrence are the strongest predictors, followed by location and crime type. Demographic features (gender, ethnicity) contribute but rank lower, consistent with the exploratory findings.

Model Comparison and Selection

The Logistic Regression model achieves a higher F1-score for the solved class, making it better suited for the primary goal: identifying cases that are likely to be solved. The confusion matrix shows it correctly identifies most solved cases but generates some false positives by classifying unsolved cases as solved. This is an expected trade-off when using class weighting on an imbalanced target.

The Random Forest model excels at identifying unsolved cases (high recall for class 0) but underperforms on the minority solved class. Despite GridSearchCV tuning, the ensemble model does not outperform logistic regression on this task, suggesting that the relationship between features and solvability may be more linear than complex.

Both models' AUC-ROC scores indicate their ability to discriminate between classes is meaningfully better than random, though the class imbalance limits the ceiling on performance metrics.

Conclusions and Recommendations

This analysis confirms that cognitive and behavioral factors around the crime itself — particularly weapon usage, crime type and location based on latitude and longitude— are the most reliable predictors of case solvability in the LAPD dataset. Victim demographics and temporal variables play secondary roles.

Based on the findings, the following recommendations are made:

- **Resource allocation:** Law enforcement should prioritize investigative resources toward weapon-involved crimes, particularly those involving bladed weapons, where solve rates are highest and additional effort is most likely to yield results
- **Data collection improvement:** The most significant limitation of both models is the class imbalance where more unsolved cases prevail. Collecting more complete case follow-up data for solved cases would directly improve model performance
- **Alternative more complex models:** XGBoost or LightGBM could potentially handle this class imbalance more effectively than standard Logistic Regression or Random Forest.
- **Geographic targeting:** Although no single area stands out dramatically, areas with below-average solve rates could benefit from targeted community policing or investigative support
- **Unknown victim data:** Cases with unknown gender or age are almost never solved. Better data capture at the point of reporting could improve both records quality and model performance